

Cloud-Native SIG present

Kubernetes for Research Infrastructure: Core Concepts and Practical Use Cases

20 August 2026, ACIT Hub Webinar

Agenda

1. Introductions
2. Introduction to Kubernetes
3. Use case 1: Lab in the Cloud
4. Use case 2: Jupyter Notebooks
5. Use case 3: Argo CD and Workflows
6. Summary
7. Q&A

Introductions

Who We Are

This webinar is run by the **Cloud-Native Special Interest Group**, a new community of RSEs/dRTPs for cloud-native technologies in research software and digital infrastructure.

Get involved at <https://cloudnative-sig.ac.uk/>

Current committee includes; Laura Shemilt¹, Alex Lubbock², Piper Fowler-Wright², Lewis Sampson³, and Tibor Auer⁴.

We are joined today by Dr Christopher Green³, Sys Admin in Scientific Computing Department

¹BioFAIR, ²Rosalind Franklin Institute, ³UKRI STFC DAFNI, ⁴University College London

The SIG so far

We have been;

- Building networks through conference attendance
- Running Kubernetes introduction workshops for RSE/dRTPs
- Sharing knowledge of Kubernetes in the DRI community through CAKE fellowship
- Formalising as a RSE Soc

If you are interested

Our workshops are available via our GitHub pages

- [Deploying a Webservice with Kubernetes](#)
- [Pocket sized Kubernetes](#): For this one, to recreate with your own hardware, see the [extra reading section](#).

You can join the CNSIG

- We are looking to grow the SIG so please join our Jisc mailing list for future updates.
 - cloudnative-sig@jiscmail.ac.uk
 - cloudnative-sig.ac.uk
- Or if you are deeply interested Kubernetes and Cloud-Native technologies join our committee. Contact any of us on email or LinkedIn to show interest

Introduction to Kubernetes

What is Kubernetes (k8s)

Kubernetes is an open-source container orchestration platform to automate the deployment, scaling & management of containerised applications across groups of machines.

Kubernetes vs Docker Compose

- **Docker:** run apps from built images in containers (sandboxed environments)
- **Docker Compose:** groups of containers with networking and storage on a single host
- **Kubernetes:** Orchestration pipeline for multi-host, multi-container applications.

Kubernetes vs Docker Compose

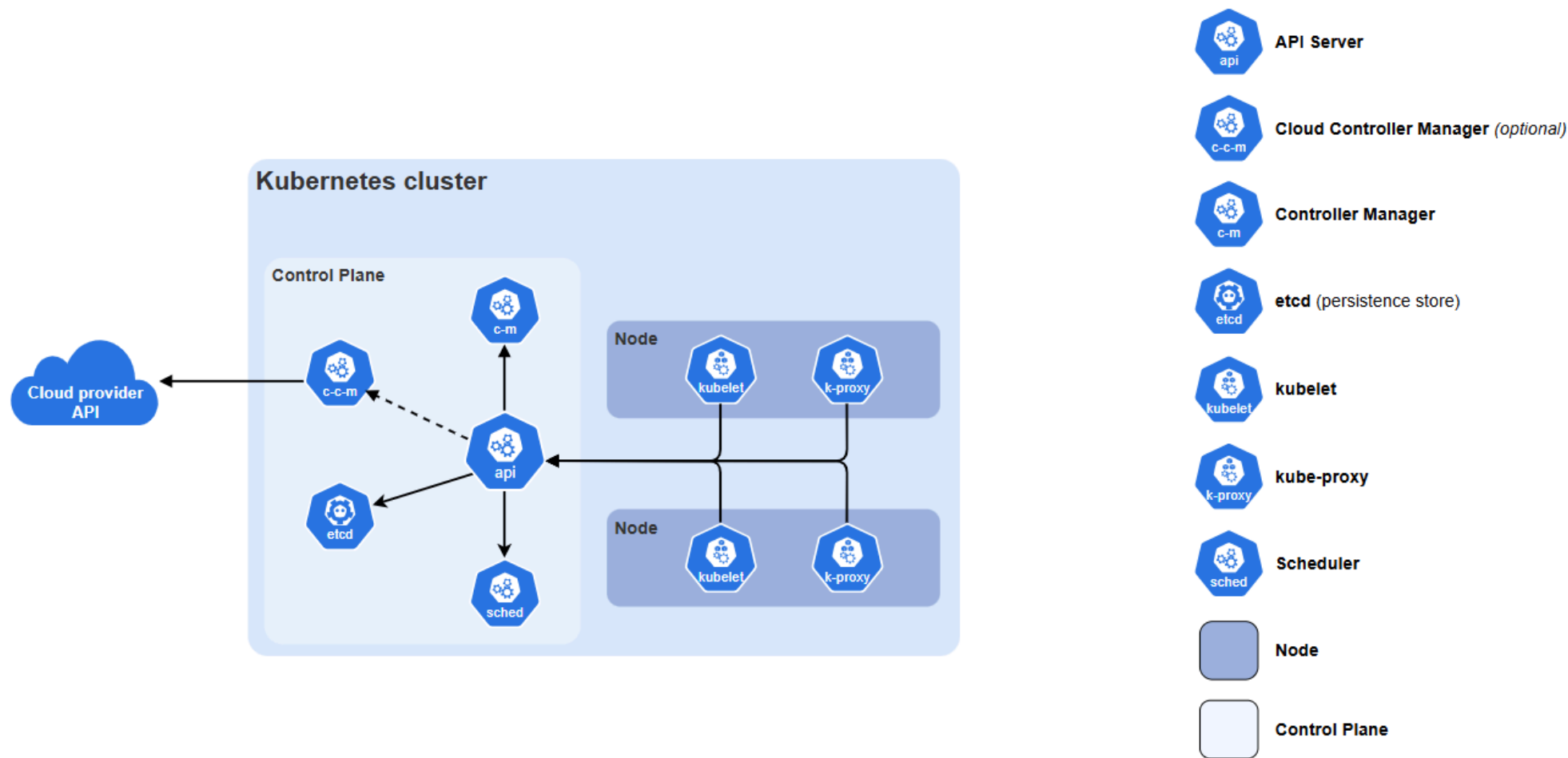
- **Docker:** run apps from built images in containers (sandboxed environments)
- **Docker Compose:** groups of containers with networking and storage on a single host
- **Kubernetes:** Orchestration pipeline for multi-host, multi-container applications.

K8s allows complex containerised applications to run across multiple hosts in a cluster with powerful automation and management features.

Key Components

- **Node:** Physical or virtual machine
- **Control Plane:** Brains of cluster, acts based on current cluster state vs target state
- **Worker:** Run containerised applications using a container runtime.
- **Pod:** Smallest deployable unit—one or more containers sharing storage/network
- **Deployment:** Target state of identical pods serving an app
- **Service:** Network endpoint for pods

Architecture Overview



Components of a Kubernetes cluster (kubernetes.io)

How Kubernetes Works

Benefits

- **Declarative** Configuration. **GitOps**: Infrastructure as code, workflow managers, e.g., [Flux](#), [ArgoCD](#)
- General **resource management** including networking and security
- Ability to **scale clusters** to include additional cloud resources.
- Offer High Availability (HA) with **Self-Healing**.

Example: On *node failure* Kubernetes distributes workload to other nodes

How Kubernetes Works

Difficulties

- Investing in an entirely new infrastructure
 - Research and Investigation
 - Migration
 - Upskilling
- Fast paced update cycle and dependency chains

Kubernetes uses in Research Computing

Use if you need to manage multiple containerised services across different machines, or for:

- Automatic workload scaling based on demand
- Rolling updates with minimal downtime
- Self-healing resilient systems with high-availability
- Portability between on-prem and cloud infrastructure

Kubernetes not uses in Research Computing

Adds complexity and not typically suitable for:

- Simple container applications on a single host (use Docker Compose)
- HPC job management (use SLURM)
- Large-scale parallel filesystems (e.g., Lustre)
- Services offered by a cloud provider or technology you are already invested in

Use cases

Wrap-up

Cloud Native SIG

This workshop was brought to you by the Cloud-Native SIG and CAKE, with support from the Software Sustainability Institute

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**Thank you for listening, any
questions?**